Solution:
Notice that the sum we're looking for is just 3 added together, several times — it does not involve i.

How many times are we adding 3? It's tempting to think it's 5 times, since $9 - 4 = 5$, but that is one short. Count out the numbers from 4 to 9 to see that the range includes 6 numbers, when we include both endpoints.

So, the sum is:

$$\sum_{i=4}^{9} 3 = 3 + 3 + 3 + 3 + 3 + 3 = 3 \cdot (9 - 4 + 1) = 3 \cdot 6 = 18$$

c) $\displaystyle\sum_{j=1}^{20} (1 - 3j)^2$

Solution:
We'll have to expand a fair bit here:

$$\begin{aligned}\sum_{j=1}^{20} (1 - 3j)^2 &= \sum_{j=1}^{20} (1 - 6j + 9j^2) \\ &= \sum_{j=1}^{20} 1 - \sum_{j=1}^{20} 6j + \sum_{j=1}^{20} 9j^2 \\ &= 20 - 6 \sum_{j=1}^{20} j + 9 \sum_{j=1}^{20} j^2\end{aligned}$$

We know that $\displaystyle\sum_{j=1}^{20} j = \frac{20 \cdot 21}{2} = 210$ and $\displaystyle\sum_{j=1}^{20} j^2 = \frac{20 \cdot 21 \cdot 41}{6} = 2870$. So, we have:

$$\begin{aligned}\sum_{j=1}^{20} (1 - 3j)^2 &= 20 - 6 \cdot 210 + 9 \cdot 2870 \\ &= 20 - 1260 + 25830 \\ &= 24590\end{aligned}$$